

Calculating Farfields from Current Data

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1 Introduction

This code intends to compute monostatic farfields from the mesh and electric/magnetic current data. All the data structure follows that of VWT-IE-Solver.

2 Theory

Monostatic farfields are computed using reciprocity theorem. Specifically, adjacent measurement is used. The process follows:

- Place a vertical (VV) and horizontal (HH) plane wave source at the angle of incidence (θ and ϕ).
- Use reciprocity to compute monostatic farfields: surface integral of the product of surface current and fields from the source.

$$\mathbf{E}_{\text{sca}}(\theta, \phi) = -j\omega \iint_S \mathbf{J}(\mathbf{r}) \cdot \mathbf{E}_{\theta, \phi}^{\text{inc}}(\mathbf{r}) dS - jk \iint_S \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\theta, \phi}^{\text{inc}}(\mathbf{r}) dS$$

3 Usage

All the input files must be in the same working path. The input files follow:

- Mesh file (`.tri`)
- In file (`.in`)
- Current data (`.curJ`, `.curM`)

e.g.,

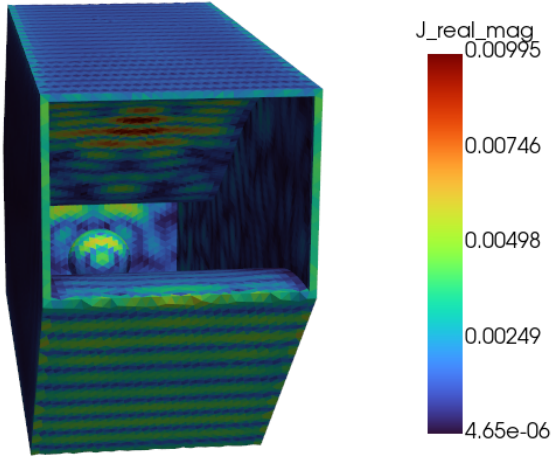
```
1 NAME.tri
2 NAME.in
3 NAME_Inc#1.curJ
4 NAME_Inc#1.curM
5 NAME_Inc#2.curJ
6 NAME_Inc#2.curM
7 ...
```

And then run the code:

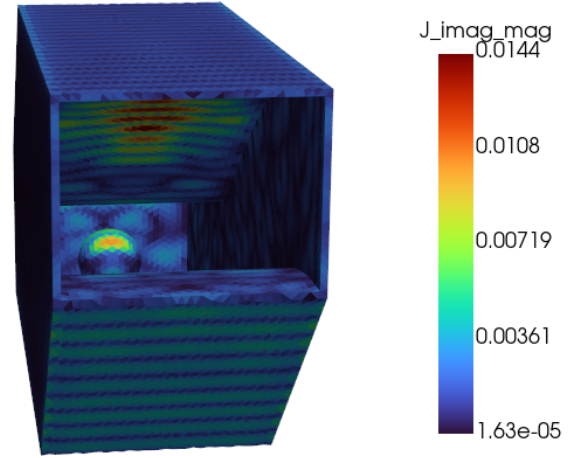
```
1 calc_efar.x NAME
```

where NAME is the name of the input files.

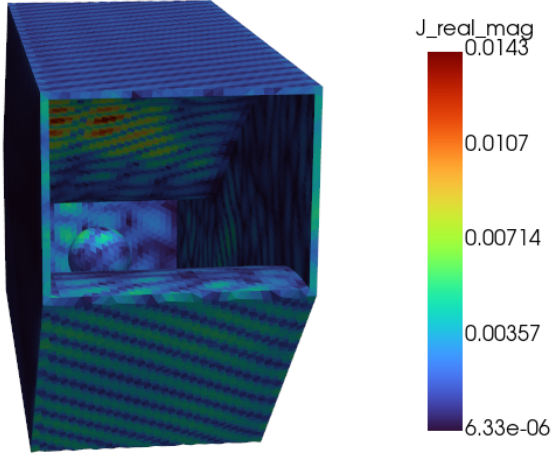
4 Example: Cobra Duct cavity



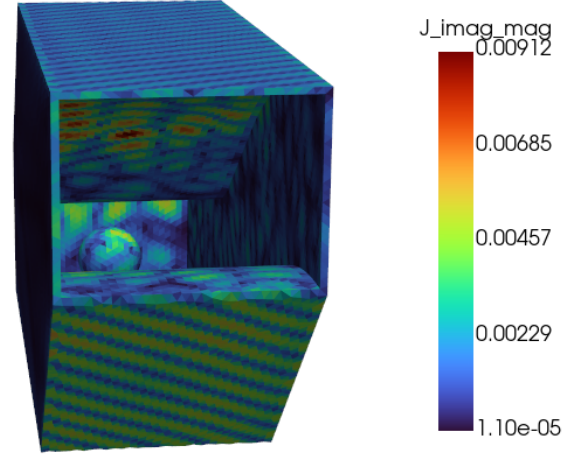
(a) J_{real} at $\phi = 0$ DEG



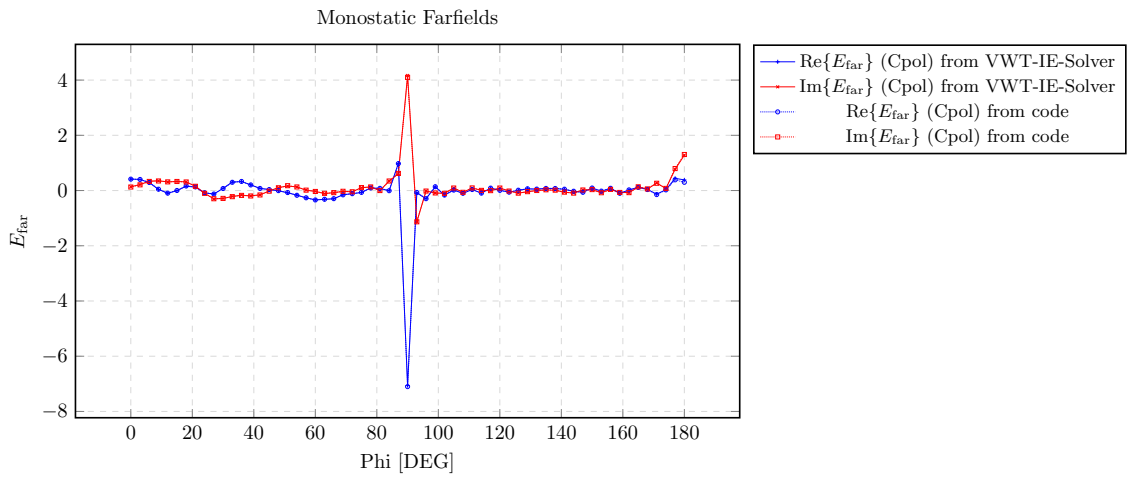
(b) J_{imag} at $\phi = 0$ DEG



(c) J_{real} at $\phi = 30$ DEG



(d) J_{imag} at $\phi = 30$ DEG



(e) Comparison of EFAR from VWT-IE-Solver and from calc.efar